

An Introduction to Hadoop and Big Data Analysis

Hadoop has become a central platform to store big data through its Hadoop Distributed File System (HDFS) as well as to run analytics on this stored big data using its MapReduce component. This article explores the basics of Hadoop.



Many of us would have certainly heard about Big Data, Hadoop and analytics. The industry is now focused primarily on them and Gartner identifies 'strategic big data' and 'actionable analytics' as being among the Top 10 strategic technology trends of 2013.

According to the Gartner website: "Big Data is moving from a focus on individual projects to an influence on enterprises' strategic information architecture. Dealing with data volume, variety, velocity and complexity is forcing changes to many traditional approaches. This realisation is leading organisations to abandon the concept of a single enterprise data warehouse containing all information needed for decisions. Instead, they are moving towards multiple systems, including content management, data warehouses, data marts and specialised file systems tied together with data services and metadata, which will become the 'logical' enterprise data warehouse."

There are various systems available for big data processing and analytics, alternatives to Hadoop such as HPC or the newly launched Red Shift by Amazon. However, the success of Hadoop can be gauged by the number of Hadoop distributions available from different technological companies such as IBM InfoSphere BigInsights, Microsoft HDInsight Service on

Azure, Cloudera Hadoop, Yahoo's distribution of Hadoop, and many more. There are basically four reasons behind its success:

1. It's an open source project.
2. It can be used in numerous domains.
3. It has a lot of scope for improvement with respect to fault tolerance, availability and file systems.
4. One can write Hadoop jobs in SQL like Hive, Pig, Jaql, etc, instead of using the complex MapReduce.

This enables companies to modify the Hadoop core or any of its distributions to adapt to the company's own requirements and the project's requirements. In this article, we will only focus on the basics of Hadoop. However, in forthcoming articles in this series, we will primarily focus on fault tolerance and the availability features of Hadoop.

Formally, Hadoop is an open source, large scale, batch data processing, distributed computing framework for big data storage and analytics. It facilitates scalability and takes care of detecting and handling failures. Hadoop ensures high availability of data by creating multiple copies of the data in different nodes throughout the cluster. By default, the replication factor is set to 3. In Hadoop, the code is moved to the location of the data instead of moving the data towards the code. In the rest of this